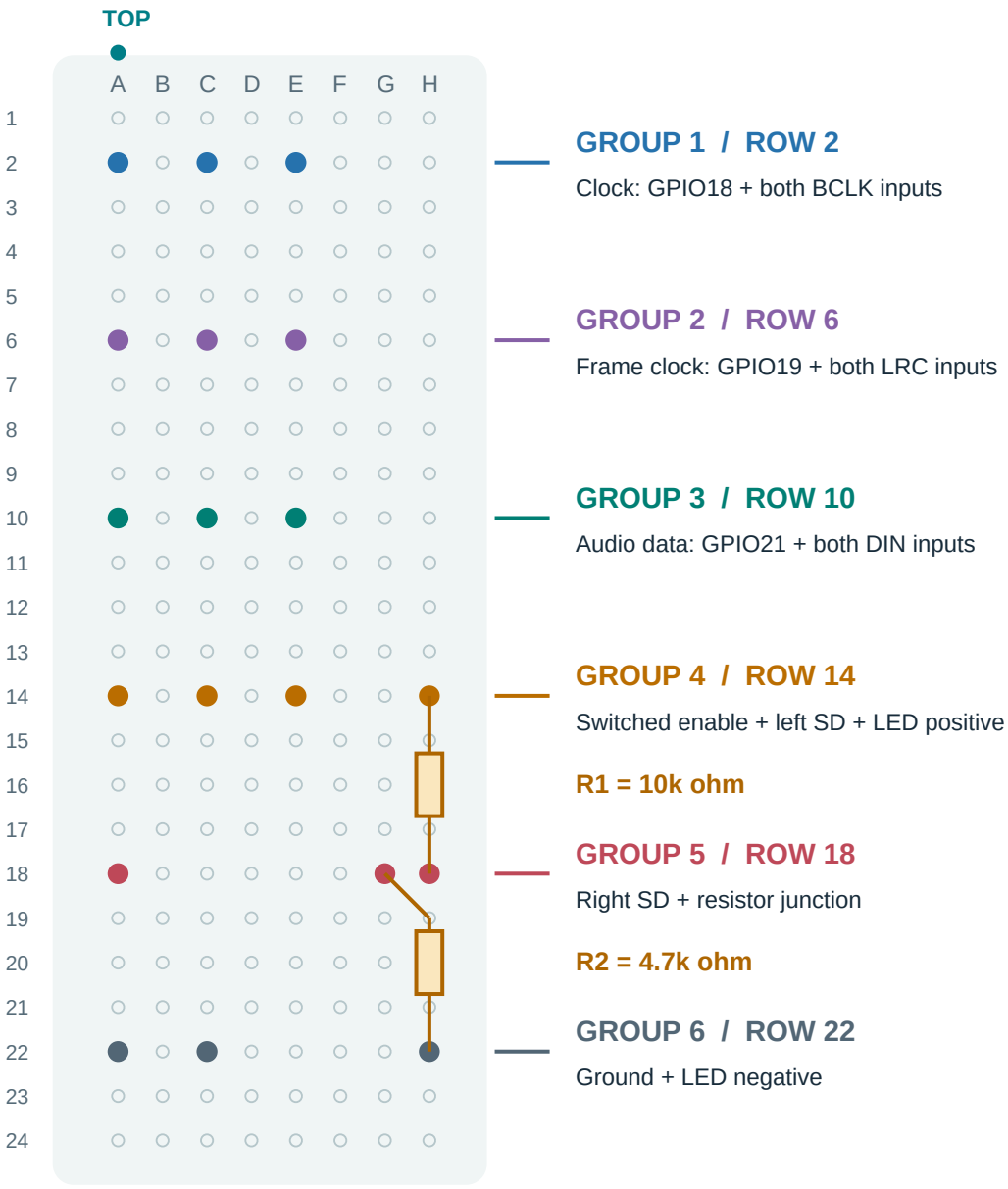
Component-side layout

50 x 70 mm individual-pad perfboard / six separate connection groups

Orientation comes first

Use 24 rows vertically. Mark TOP and a dot at the top-left on the component side.
Coordinates below are our own: row 1 at the top; columns A-H left to right.
Use eight consecutive columns. Ignore conflicting printed labels; leave other holes unused.



Three empty rows between each group

Rows 3-5, 7-9, 11-13, 15-17 and 19-21 remain empty. Diagram is not to scale.
Each coloured dot is a connection position; the exact assignments are on page 2.
R1 ends at H18; R2 starts at G18. The row-18 bus joins those two pads.
R2 runs diagonally from G18 to H22. Resistors have no polarity.

Connection schedule

Transfer one labelled wire at a time / GPIO numbers are BCM, not physical pin numbers

GROUP 1 / row 2

A2	GPIO18 - yellow / MALE
C2	LEFT amp BCLK - brown / FEMALE
E2	RIGHT amp BCLK - orange / FEMALE

GROUP 2 / row 6

A6	GPIO19 - blue / MALE
C6	LEFT amp LRC - red / FEMALE
E6	RIGHT amp LRC - yellow / FEMALE

GROUP 3 / row 10

A10	GPIO21 - green / MALE
C10	LEFT amp DIN - purple / FEMALE
E10	RIGHT amp DIN - blue / FEMALE

GROUP 4 / row 14

A14	Toggle WHITE (COM) - GREEN extension
C14	LEFT amp SD - white / FEMALE
E14	Toggle YELLOW (LED+) - YELLOW extension
H14	R1 (10k) first end

GROUP 5 / row 18

A18	RIGHT amp SD - green / FEMALE
H18	R1 (10k) second end
G18	R2 (4.7k) first end; use G18 to avoid sharing a hole

GROUP 6 / row 22

A22	Toggle BLUE (LED-) - BLUE extension
C22	GPIO GND / MALE
H22	R2 (4.7k) second end

Toggle supply wires - updated after your swap

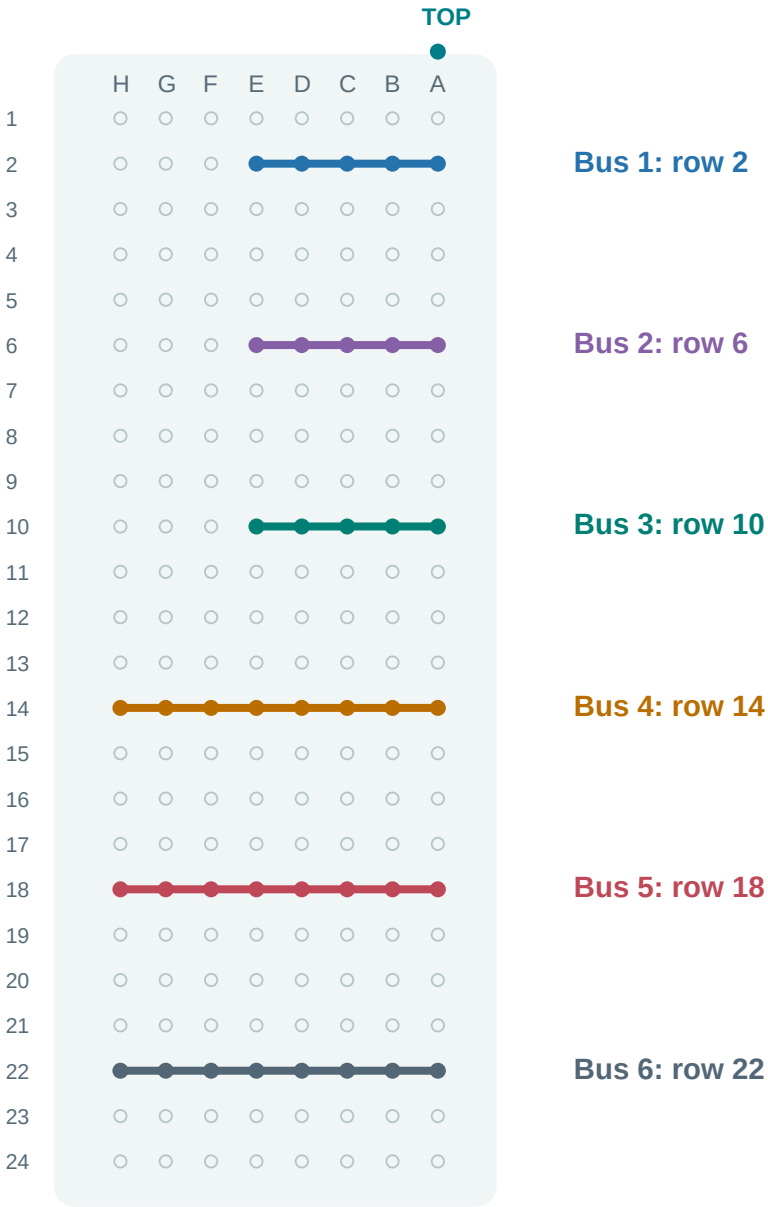
Toggle BLACK (NO) -> GND. Toggle ORANGE (NC) -> 3.3 V.
Both use MALE ends at the breakout; follow native labels, not extension colours.
Amplifier VIN and amplifier GND supply wiring remain as already connected.

Solder-side wiring

Flip the board left-to-right, keeping TOP at the top / the columns are mirrored

Create the connections yourself

This board has isolated pads: holes in a row are NOT connected automatically.
Run a solid-wire bus along each occupied row and solder it to each used pad.
Use A-E for groups 1-3, A-H for groups 4-6. Never join one bus to another.



Before reconnecting power

1. Disconnect external power and the UPS battery supply before soldering.
2. Check continuity along each bus; inspect for solder bridges between buses.
3. Check resistor values with external wiring disconnected if readings are unclear.
4. Secure cables against pulling, then verify SD-to-GND voltage in both switch positions.
5. Mount on insulating standoffs or a base; keep the exposed solder side off metal.

Right-amp fault resolved: user confirmed contact with a conductive surface caused it.

Hole & connector labels

Component-side view / connector type refers to the free end of the cable

Solder the stripped wire end into the perfboard hole

FEMALE sockets fit the amplifier header pins. MALE pins connect to your GPIO breakout.
LEFT / RIGHT identify the speaker amplifier. Speaker leads use its screw terminals.

ROW 2

A2
GPIO18
Breakout
MALE

C2
LEFT speaker amp
BCLK
FEMALE

E2
RIGHT speaker amp
BCLK
FEMALE

ROW 6

A6
GPIO19
Breakout
MALE

C6
LEFT speaker amp
LRC
FEMALE

E6
RIGHT speaker amp
LRC
FEMALE

ROW 10

A10
GPIO21
Breakout
MALE

C10
LEFT speaker amp
DIN
FEMALE

E10
RIGHT speaker amp
DIN
FEMALE

ROW 14

A14
Toggle WHITE / COM
GREEN extension
Solder wire

C14
LEFT speaker amp
SD
FEMALE

E14
Toggle YELLOW / LED+
YELLOW extension
Solder wire

H14
R1 / 10k
First end
Resistor lead

ROW 18

A18
RIGHT speaker amp
SD
FEMALE

G18
R2 / 4.7k
First end
Resistor lead

H18
R1 / 10k
Second end
Resistor lead

ROW 22

A22
Toggle BLUE / LED-
BLUE extension
Solder wire

C22
GPIO GND
Breakout
MALE

H22
R2 / 4.7k
Second end
Resistor lead

Toggle labels show native switch colour first, then the extension colour at the perfboard.
BLACK -> GND; ORANGE -> 3.3 V at the breakout (MALE ends, after your swap).
Unlisted holes have no component or cable. Bus wires join the pads shown on page 3.